
Intelligence Engineering

Solving Real World Problems At Scale

Prof. Dr.-Ing. Mark Schutera | September 25, 2026 · *drifting stream*

LECTURE VII

Model Deployment and Prediction Service

This lecture in one slide

VII Model Deployment and Prediction Service

Machine Learning Deployment Myths

Batch Prediction Versus Online Prediction

Model Compression

ML on the Cloud and on the Edge

*why this lecture matters, and what
you can do after it*

1

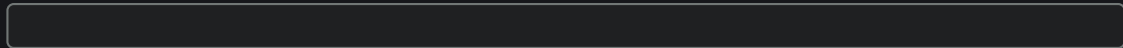
Machine Learning Deployment Myths

Myth 1: You Only Deploy One or Two ML Models at a Time

$P(y|x)$

[1]

Myth 2: If We Don't Do Anything, Model Performance Remains the Same



Myth 3: You Won't Need to Update Your Models as Much

$P(y | x)$

[1]

Myth 4: Most ML Engineers Don't Need to Worry About Scale

Definition: Scale

2

Batch Prediction Versus Online Prediction

From Batch Prediction to Online Prediction

$P(y|x)$

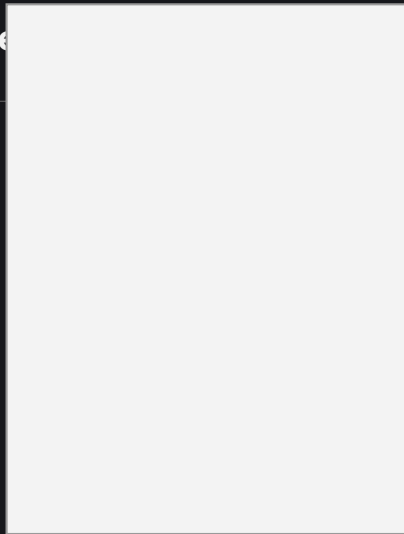
Batch prediction

Online prediction

[1]

Unifying Batch Pipeline and Streaming Pipeline

Batch pipeline and streaming pipeline



[1]

$P(y | x)$

3

Model Compression

Low-Rank Factorization

$P(y|x)$

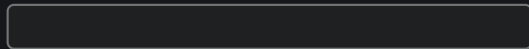
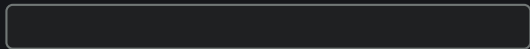
[1]

Knowledge Distillation

Definition: Knowledge distillation

Pruning

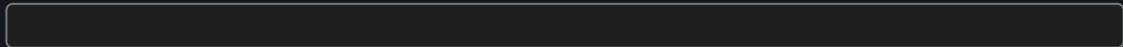
$$P(\hat{y} | x)$$



$$[1]$$

Quantization

$P(y|x)$



[1]

4

ML on the Cloud and on the Edge

Compiling and Optimizing Models for Edge Devices

Compiling

Optimizing

Model optimization

Model optimization

$P(y | x)$

[1]

Using ML to optimize ML models

$P(\hat{y} | x)$

[1]

ML in Browsers

Browser

5

Summary

Summary

$P(y|x)$

[1]

References

- 1 Chip Huyen. *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications*. Sebastopol, CA: O'Reilly Media, 2022.